to exist, this infinite, homogeneous substance would completely fill the universe, and therefore, the great many varieties of phenomena would remain unexplained, and for this reason, Change and Becoming must have arisen from that indeterminate prime basis of all things. Anaximander seems to have regarded the process of Becoming as some sort of degeneration or debasement of this undifferentiated Being—as an escape, as it were, ultimately expiated by a return into that which is without shape or character.

In the philosophy of Heraclitus, the concept of Becoming occupies the foremost place. He regarded that which moves—fire—as the basic element. In the teachings of Parmenides, a fundamental polarity—that of Being and Not-Being—is the central concept. Parmenides, too, regarded the wide variety of phenomena as resulting from the combined action and reaction of two opposed principles.

Anaxagoras, who followed Thales by about a century (he probably lived about 500 B.C.), was responsible for a definite transition to a more materialistic view of the world of phenomena. He assumed that there existed an infinite number of basic substances, the mutual interactions of which produced the variety of world processes. In his view these basic substances possessed the character of purely material elements in a much greater degree; he conceived of them as being eternal and indestructible in themselves, and he considered that the change and sequence of phenomena were produced solely by their sharing in the movement which threw them together at random.

Empedocles, about ten years later, saw the existence of four ‘elements’—earth, air, fire and water—as the ‘prime root’ of all things. He regarded the primordial state of all things as consisting in an undifferentiated, homogeneous mixture of the elements, bound by Love in a state of eternal bliss, whereas Hate tended to separate these elements and to shape out of them the variegated drama of Life.

This pronounced tendency to materialism reached its highest development with the philosophers Leucippus, a contemporary of Empedocles, and Democritus who was Leucippus’ pupil. The antithesis of Being and Not-Being became crystallized in the doctrines of Leucippus as the antithesis of ‘Full’ and ‘Empty’. The concept represented by ‘Full’ was regarded as manifesting itself in the ultimate, indivisible particles, the *atoms*, between which there was nothing but emptiness. The atom was pure Being, eternal and indestructible, but inasmuch as there existed an infinite number of atoms, pure Being could, within certain limits, be repeated an indefinite number of times. Thus, for the first time in history, there was voiced the idea of the existence of smallest ultimate, indivisible particles—the atoms—as the fundamental building blocks of all matter. In this manner, the concept of matter became analysed, in fact, into two sub-concepts: atoms and the void in which the atoms move. Formerly, space had seemed to be filled by matter; it was, as it